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Course : Software Engineering

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Experiment – 24

Question :

Create a CI/CD Pipeline Using GitHub Actions

Tasks:

- Set up a GitHub repository for a containerized Flask application.
- Define a GitHub Actions workflow file to automate testing and deployment.
- Add steps to build, test, and push Docker images to Docker Hub.
- Deploy the application to a cloud platform (Heroku/AWS).
- Submit the workflow file and deployment link.

Submission :

GitHub Repository: <https://github.com/tamilelakiya07/flask-cicd.git>

Docker Hub: <https://hub.docker.com/repository/docker/tamilelakiya19/flask-cicd/general>

Workflow File: github/workflows/ci-cd.yml

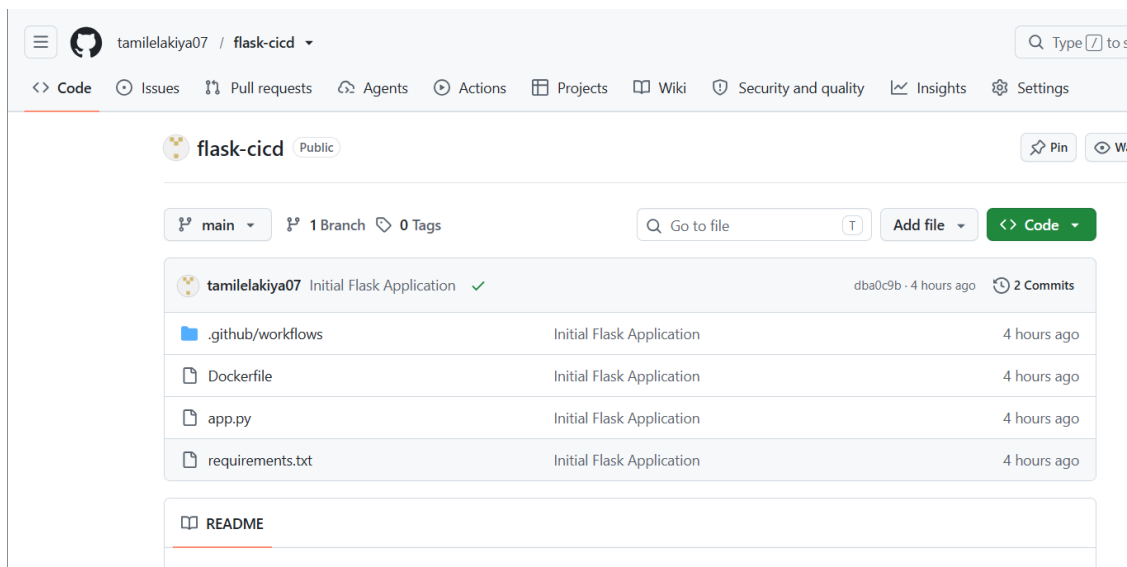
Deployment Link: <https://flask-cicd-onxb.onrender.com>

Aim

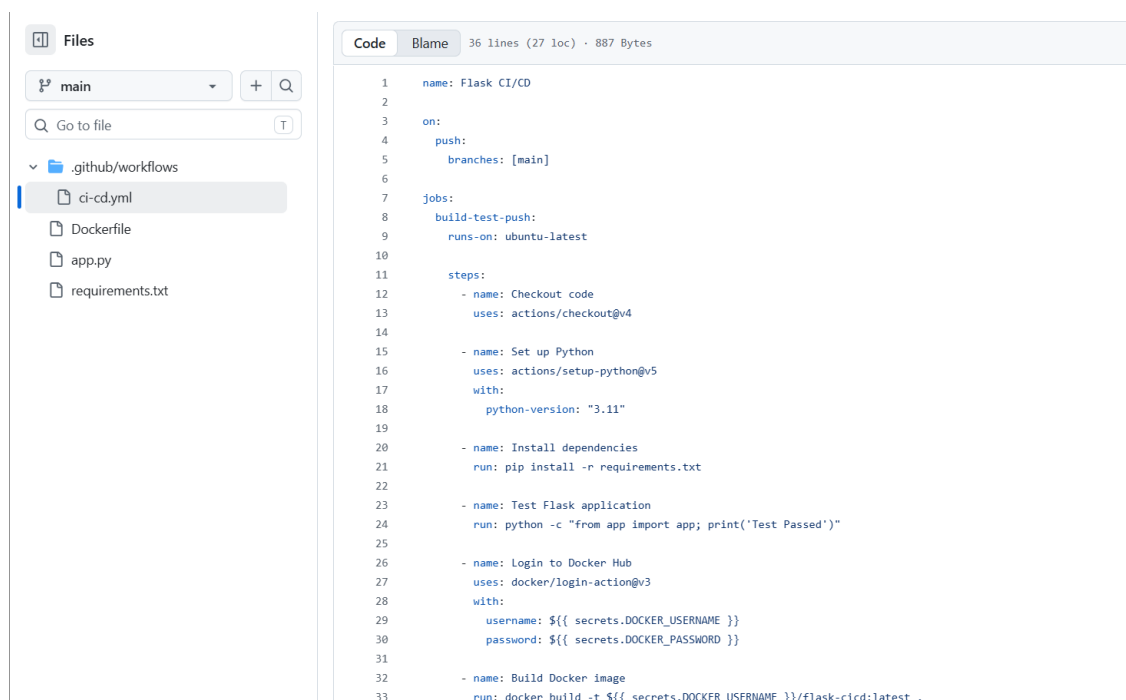
To create a CI/CD pipeline for a containerized Flask application using GitHub Actions, Docker Hub, and a cloud platform.

Procedure

1. Created a Flask application using app.py and requirements.txt.
2. Created a Dockerfile to containerize the Flask application.
3. Tested the Flask application locally.
4. Created a GitHub repository and pushed the project.



5. Created the GitHub Actions workflow file: .github/workflows/ci-cd.yml

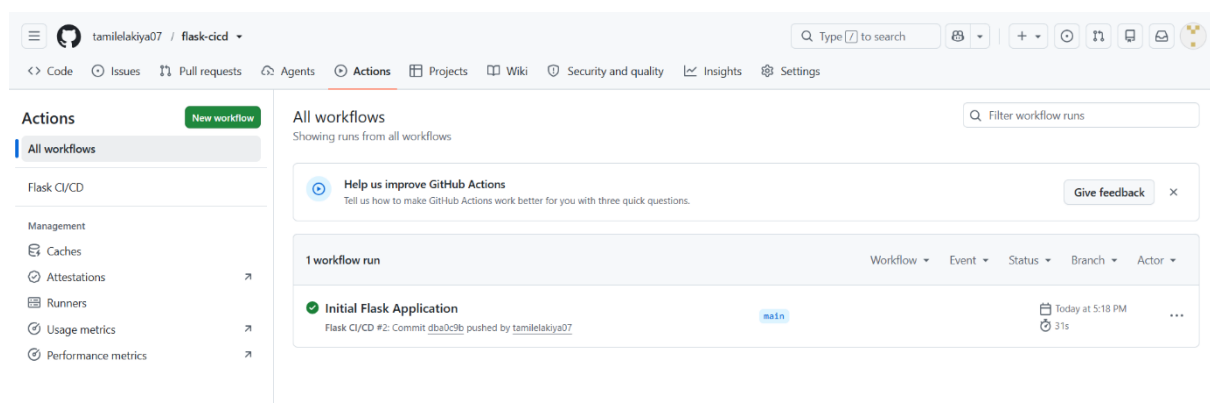


6. Configured GitHub Actions to:

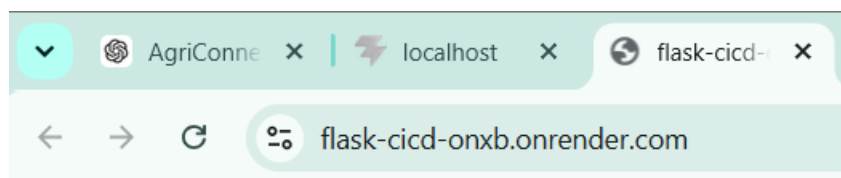
- Checkout the source code
- Install Python dependencies
- Test the Flask application
- Build the Docker image
- Login to Docker Hub
- Push the Docker image to Docker Hub

7. Added DOCKER_USERNAME and DOCKER_PASSWORD as GitHub repository secrets.

8. Verified the workflow execution under the **Actions** tab.



9. Deployed the Dockerized Flask application to a cloud platform.



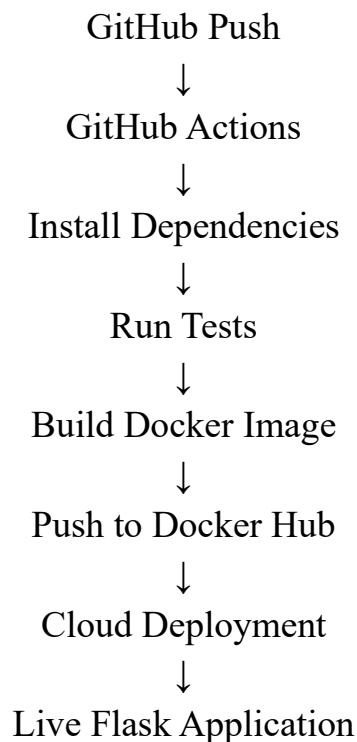
Hello! Flask CI/CD Pipeline is Working.

10.Verified the application using the generated deployment URL.

The screenshot displays the OnRender dashboard for a project named 'flask-cicd'. The left sidebar contains navigation options: Environment, flask-cicd, Events (selected), Settings, MONITOR, Logs, Metrics, and MANAGE. A notification banner at the top states: 'Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.' Below this, the deployment status is shown as 'August 28, 2026 at 9:45 PM' with a 'Live' indicator. The deployment name is 'dba0c9b Initial Flask Application'. The logs section shows the following output:

```
Aug 28
09:46:28 PM [2qswx] * Running on all addresses (0.0.0.0)
09:46:28 PM [2qswx] * Running on http://127.0.0.1:5000
09:46:28 PM [2qswx] * Running on http://10.26.21.60:5000
09:46:28 PM [2qswx] Press CTRL+C to quit
09:46:31 PM [2qswx] 127.0.0.1 - - [28/Aug/2026 16:16:31] "HEAD / HTTP/1.1" 200 -
09:46:35 PM [2qswx] ==> Your service is live 🎉
09:46:35 PM [2qswx] 127.0.0.1 - - [28/Aug/2026 16:16:35] "GET / HTTP/1.1" 200 -
09:46:35 PM [2qswx] ==> Available at your primary URL https://flask-cicd-onxb.onrender.com
09:46:44 PM [2qswx] 127.0.0.1 - - [28/Aug/2026 16:16:44] "GET / HTTP/1.1" 200 -
09:46:45 PM [2qswx] 127.0.0.1 - - [28/Aug/2026 16:16:45] "GET /favicon.ico HTTP/1.1" 404 -
09:47:27 PM [2qswx] ==> Detected a new open port HTTP:5000
```

CI/CD Workflow



Result

The CI/CD pipeline was successfully created. GitHub Actions automatically tested the Flask application, built and pushed the Docker image to Docker Hub, and the application was deployed to the cloud.